

Response Letter with Annotations

Dash Nelson, G., & Rae, A. (2016). An economic geography of the United States: From commutes to megaregions. *PLoS ONE*, 11(11), e0166083.

<https://doi.org/10.1371/journal.pone.0166083>

Annotations

1. Purposes

- a. Motivation (one-sentence summary): The study uses cartographic techniques in order to provide a rigorous and evidence-based spatial clustering of urban “megaregions”.
- b. Specific Rationale (several bullet points)
 - Traditional regional boundaries are often subjective and administratively defined.
 - Commuting flows reflect real economic and social connections.
 - Existing approaches struggle with empirical problem of defining clear, objective boundaries.
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 - IV: commuting flows
 - DV: megaregions regionalization

2. Method

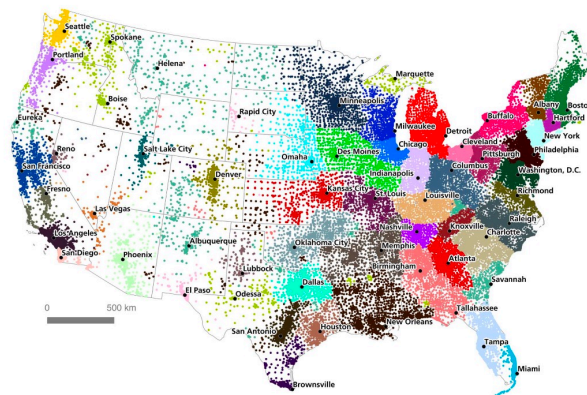
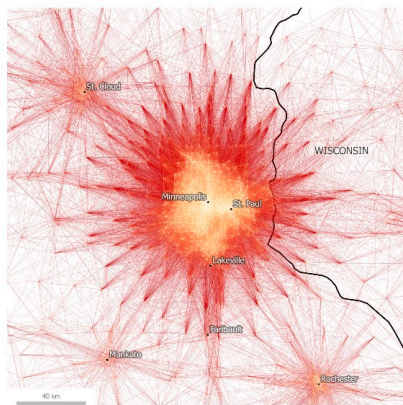
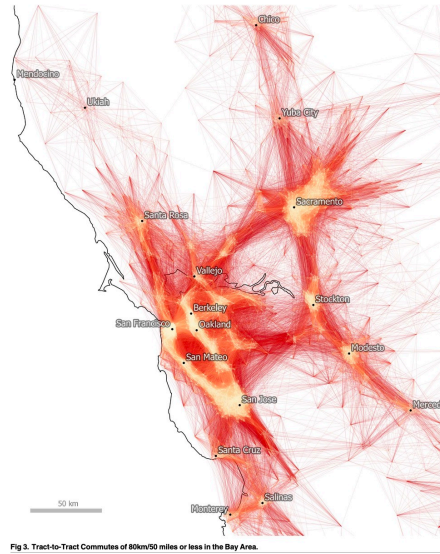
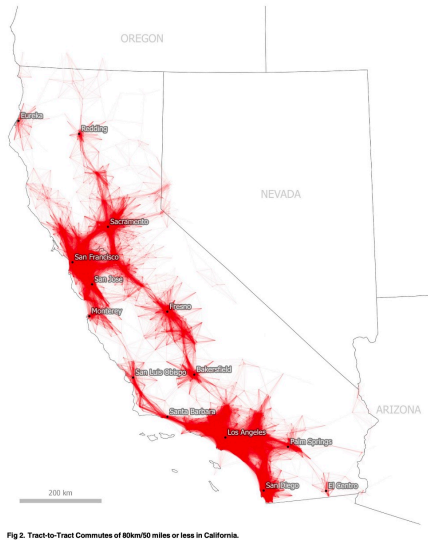
- a. Research Design
 - A visual heuristic approach:
 - Visualize large-scale commuting networks to identify megaregions using maps of nodes and edges
 - An algorithmic approach:
 - Use Combo algorithm to optimizes modularity to identify communities
- b. Sample (representative sample?) over 130 million commuters, 74,000 census tracts
- c. Measures
 - **Nodes:** Census tracts
 - **Edges:** Commuting flows
 - **Distance:** Euclidean distance (used for filtering)
 - **Modularity:** the total number of connections within a network set which lie entirely within a proposed community boundary
- d. Analysis:
 - Apply community detection algorithm (Combo) to construct network from commuting data
 - Compare results with real geographic regions

3. Results and Conclusions

- a. Answers to RQs and Hypotheses

- Communities are internally dense, externally sparse with commuting networks
- Communities appear as coherent spatial regions
- Some regions are highly interconnected (e.g., LA–San Diego) while some are Self-contained (e.g., Minneapolis)

b. Include Key Figures/Tables



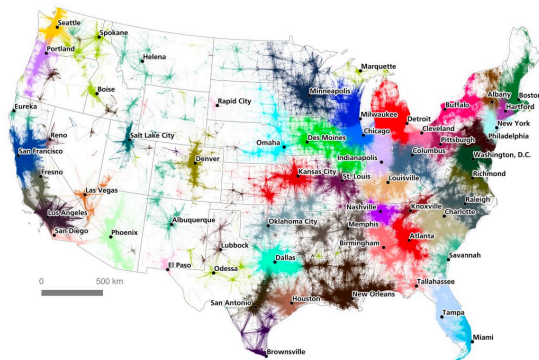


Fig 6. A commuter flow-based regionalization of the United States.
doi:10.1371/journal.pone.0166083.g006

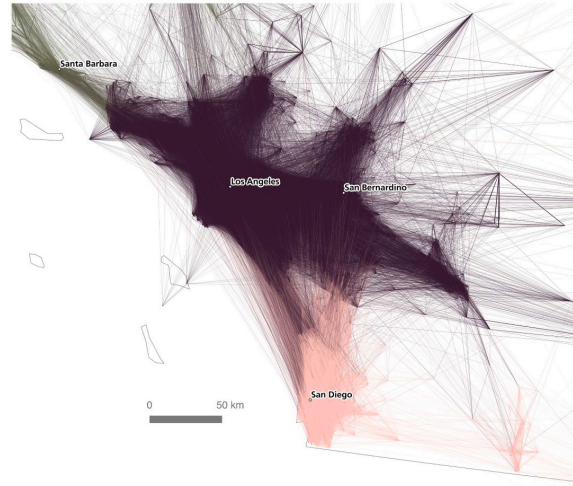


Fig 7. Relative density of connections between the Los Angeles and San Diego regions.
doi:10.1371/journal.pone.0166083.g007

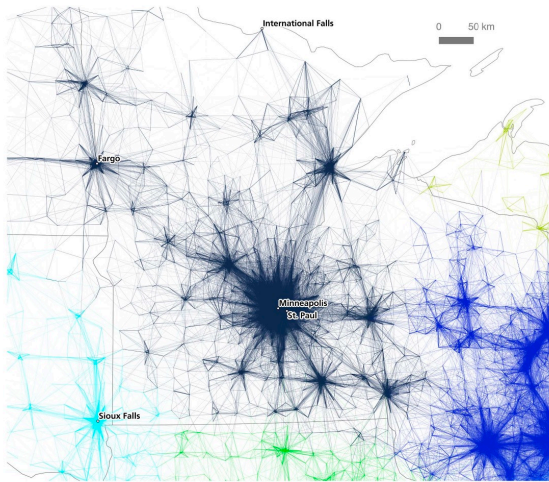


Fig 8. The Twin Cities region.
doi:10.1371/journal.pone.0166083.g008

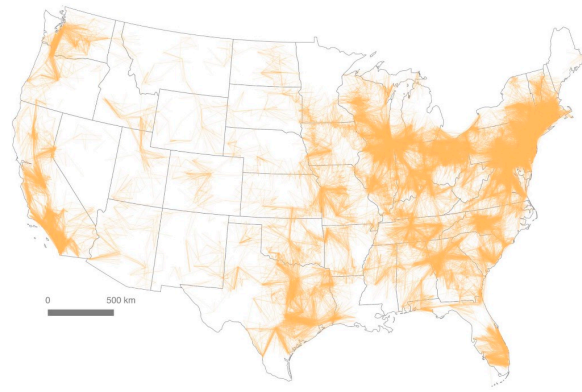


Fig 9. Inter-community flows for the United States.
doi:10.1371/journal.pone.0166083.g009

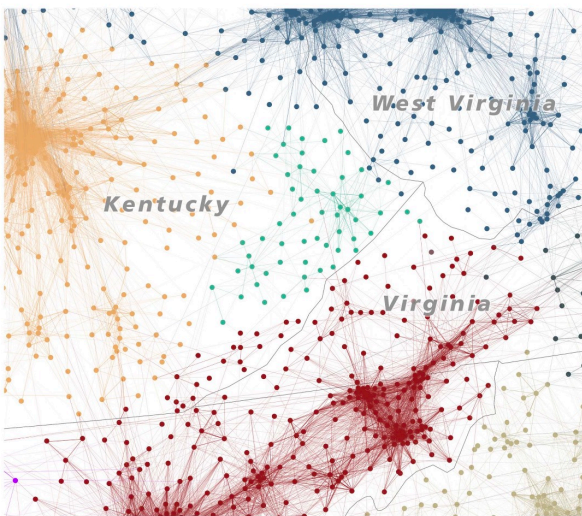


Fig 10. The Eastern Kentucky isolate region.
doi:10.1371/journal.pone.0166083.g010



Fig 11. Computed communities subject to interpretive analysis.
doi:10.1371/journal.pone.0166083.g011

c. Authors' Conclusions

- U.S. geography can be understood as a system of functional regions

- Regions are emergent from human behavior
- Megaregions are real but not fixed or unique
- Regional boundaries are fuzzy rather than discrete

4. Discussion

a. Implications

- The algorithmic approach provides a new, detailed view of the economic geography of the United States, which can be of potential use in a wide variety of fields, including transit planning, real estate, retail analysis, and infrastructure development, to name just a few.

b. Limitations

- Uses only commuting data (ignores other flows)
- Enforces hard boundaries despite fuzzy reality

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Response Letter (~600 words)

This study provides a methodologically rigorous approach to understanding the economic geography of the United States. Rather than relying on administratively defined regions, the authors attempt to identify functional economic regions from patterns of commuting flows. By treating commuting flows as a network and applying community detection algorithms, the study shifts the focus from predefined spatial boundaries to connectivity embedded in human economic activities. I think it is the most novel contribution of the article, as it reframes regions from the dimension of economic connectivity rather than fixed administrative units.

One of the main strengths of the study lies in its integration of visual and computational approaches. The visualizations of commuting networks provide an intuitive sense of how people commute across different regions, revealing dense urban centers and corridors of interaction. Additionally, the authors go beyond descriptive visualization, using an algorithmic approach that systematically partitions the networks into communities. This is the most significant methodological innovation, as it addresses the limitation of visual interpretation, which is inherently subjective and cannot produce clear and reproducible boundaries.

The use of modularity as an optimization criterion further strengthens the methodological rigor of the study. Instead of arbitrarily defining regions, the algorithm identifies communities with maximal internal connectivity and minimal external connectivity. This provides a clear and replicable standard for evaluating community structures. Moreover, the findings that these communities are also spatially clustered, even though geographic information is not used in the computation, is particularly interesting. It suggests that commuting flows themselves are embedded in spatial relationships, offering indirect support for Tobler's First Law that geographically nearby places are more likely to be economically connected.

For the visualization, I really appreciate these amazing figures, which gives a visually appealing illustration of connectivity patterns within and across communities. One of the most striking aspects, from my perspective, is how the figures (e.g., Fig 5 and Fig 6) make an abstract computational result feel "real." By mapping community detection outcomes back onto geographic space, the authors successfully bridge the gap between network analysis and spatial intuition. As a reader, I found this extremely helpful in understanding how large-scale regional structures might also emerge from everyday commuting behaviors.

At the same time, the study also raises important limitations. First, the reliance on commuting data as the sole indicator of megaregion identification may be restrictive. Commuting captures only one type of economic interaction, primarily related to labor markets. Other forms of connection, such as digital communication, economic exchange, or cultural ties, are not included, which may lead to an incomplete representation of community structure in the United States.

Second, the algorithm enforces a hard partition, meaning that each census tract is assigned to only one community. While this is necessary for optimization and replicability, it may oversimplify reality. As shown in the visualizations, especially in boundary areas, some locations appear to be strongly connected to multiple regions. These transition zones suggest that regional boundaries may be inherently fuzzy and overlapping. By forcing discrete partition, the model may undermine the complexity.

For the visualization, I cannot find any faults of these graphs, but some aspects that can be easily misunderstood. For instance, in Fig 5, the density of points might intuitively be interpreted as indicators of stronger connectivity, while in fact it mainly reflects population distribution (e.g., the number of census tracts). Another point is the density of information. In figures with many overlapping lines (e.g., Fig 6), it is hard to distinguish patterns beyond the most obvious ones. While the information is all meaningful, it can also create a kind of visual overload, where some nuance structural differences are difficult to perceive.